

# Hao Xu (徐浩)

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## RESEARCH POSITIONS

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|------------------------------------------------------------------------------|--------------|
| Post doctorate, Optoelectronics, <i>Tsinghua University</i> , Beijing, China | 04/'24 – Now |
| Doctorate researcher, Optomechanics, <i>CNRS-IEMN</i> , Paris/Lille, France  | 2020 – 2024  |
| R.A., Nanophotonics, <i>Tsinghua University TBSI</i> , Shenzhen, China       | 2018 – 2020  |
| Advisor: Baofu Ding, Bilu Liu                                                |              |
| Winter school program, <i>University of California</i> , San Diego, USA      | 02-03/2017   |

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Ph.D. in Nanoelectronics, <i>Universite de Lille</i> , Lille, France                                                                                         | 2020 – 2024 |
| Thesis: <i>Analysis of device properties of silicon nitride membrane nanoelectromechanical resonators for future optomechanical thermometry applications</i> |             |
| Advisor: Didier Théron, Xin Zhou                                                                                                                             |             |
| M.Eng. in Electronics, <i>Xi'an Jiaotong University</i> , Xi'an, China                                                                                       | 2015 – 2018 |
| Advisor: Wenhui Yi                                                                                                                                           |             |
| B.Sc. in Optics, <i>Harbin University of Sci and Tech</i> , Harbin, China                                                                                    | 2010 – 2014 |

## RESEARCH INTERESTS

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His interests focus on micro and nanosystems & sensors, including

- Micro and nanofabrication (cleanroom process)
- Nanoelectromechanical device modeling (math, physics)
- Hands-on experiments (electrical, optical readout)
- In particular, semiconductor devices, neuroelectronics and signal processing

## AWARDS

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|------------------------------------------------------------------------------------------------------------------|-------------|
| Postdoctoral research fellowship, <i>Tsinghua University</i>                                                     | 2024 – 2027 |
| Doctorate scholarship, <i>Centre National de la Recherche Scientifique</i>                                       | 2020 – 2024 |
| Academic scholarship, <i>Xi'an Jiaotong University</i>                                                           | 2015 – 2018 |
| Excellent Young Volunteer Award, <i>Xi'an Jiaotong University</i>                                                | 10/2016     |
| Third prize (province) of National Undergraduate Mathematical Contest, <i>Chinese Mathematical Society (CMS)</i> | 11/2011     |

## REFERENCES

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|                                         |                             |
|-----------------------------------------|-----------------------------|
| Prof. Kaiyu Cui, Tsinghua University    | kaiyucui@tsinghua.edu.cn    |
| Prof. Yidong Huang, Tsinghua University | yidonghuang@tsinghua.edu.cn |
| Prof. Bilu Liu, Tsinghua University     | bilu.liu@sz.tsinghua.edu.cn |
| Prof. Didier Theron, CNRS & Univ. Lille | didier.theron@univ-lille.fr |
| Prof. Gaelle Lissorgues, ESIEE Paris    | gaelle.lissorgues@esiee.fr  |

## PRESENTATIONS & CONFERENCES

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| NEMS talk (Invited) at Beijing Academy of Quantum Info Sciences     | 01/2025     |
| Nano-optoelectronics group meeting at Tsinghua University, Beijing  | 09/2024     |
| Department EE talk (Invited) at Xi'an Jiaotong University, Xi'an    | 01/2024     |
| NEMS for interface science (Invited) at EPFL, Lausanne, Switzerland | 12/2023     |
| Global Young Scholars forum (Invited) at CUHK-Shenzhen, Shenzhen    | 12/2023     |
| Red wine & Cheese lunch talk at University of Bordeaux, France      | 10/2022     |
| Microsystems group meeting at CNRS-IEMN, Lille/Paris, France        | 12/2021     |
| Optics group meeting at Tsinghua University TBSI, Shenzhen          | 2018 – 2020 |
| Molecular imaging group meeting at Peking University, Beijing       | 05/2018     |
| Biomechanics group meeting at Xi'an Jiaotong University, Xi'an      | 04/2017     |
| Juice & Snacks lunch talk at University of California, San Diego    | 02/2017     |
| Electronics group meeting at Xi'an Jiaotong University, Xi'an       | 2015 – 2018 |
|                                                                     |             |
| Condensed Matter Physics seminar (Invited) at Peking University     | 11/2024     |
| Quantum Physics IAS workshop (virtual meeting) at Princeton         | 08/2024     |
| Frontiers of Nanomechanical Systems (FNS), Delft, The Netherlands   | 06/2023     |
| LIMMS-IEMN workshop on NEMS/MEMS (virtual meeting)                  | 12/2022     |
| Technologies for Neuroengineering (virtual Nature conferences)      | 10/2022     |
| Nanomechanics and optomechanics seminar, Bordeaux, France           | 10/2022     |

## PUBLICATIONS

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First-author and essential contribution papers (# Equal contribution):

[Subwavelength-scale InGaP nanolasers with low threshold for multi-channel biophotonic probes](#) (2026, Under review)

Shuyuan Li#, [Hao Xu](#)#, Yongzhuo Li\*, Hao Sun, Fang Liu, Xue Feng, Wei Zhang, Kaiyu Cui\*, Yidong Huang\*

[Imaging nanomechanical vibrations and manipulating parametric mode coupling via scanning microwave microscopy](#)

[Hao Xu](#), Srisaran Venkatachalam, Toky-Harrison Rabenimanana, Christophe Boyaval, Sophie Eliet, Flavie Braud, Eddy Collin, Didier Theron, Xin Zhou\*

(\* corresponding author)

2024, **Nano Letters**, 24 (28), 8550–8557.

[Fabrication of silicon nitride membrane nanoelectromechanical resonator](#)

[Hao Xu](#), Srisaran Venkatachalam, Christophe Boyaval, Pascal Tilmant, Francois Vaurette, Yves Deblock, Didier Theron, Xin Zhou\*

2023, **Microelectronic Engineering**, 280 (2023) 112064.

[Coupling capacitively distinct mechanical resonators for room temperature phonon-cavity electromechanics](#)

Alok Pokharel, [Hao Xu](#), Srisaran Venkatachalam, Eddy Collin, Xin Zhou\*  
2022, **Nano Letters**, 22 (18), 7351-7357.

[Magnetically tunable and stable deep-ultraviolet birefringent optics using two-dimensional hexagonal boron nitride](#)

[Hao Xu](#), Baofu Ding\*, Youan Xu, Ziyang Huang, Dahai Wei, Shaohua Chen, Tianshu Lan, Yikun Pan, Hui-Ming Cheng\* and Bilu Liu\*  
(\* corresponding author)  
2022, **Nature Nanotechnology**, 17, 1091-1096.

[A 2D material-based transparent hydrogel with engineerable interference colours](#)

Baofu Ding, Pengyuan Zeng, Ziyang Huang, Lixin Dai, Tianshu Lan, [Hao Xu](#), Yikun Pan, Yuting Luo, Qiangmin Yu, Hui-Ming Cheng and Bilu Liu\*  
2022, **Nature Communications**, 13(1), 1-8.

[High-q silicon nitride drum resonators strongly coupled to gates](#)

Xin Zhou\*, Srisaran Venkatachalam, Ronghua Zhou, [Hao Xu](#), Alok Pokharel, Andrew Fefferman, Mohammed Zaknoute, Eddy Collin  
2021, **Nano Letters**, 21(13), 5738-5744.

[Obtaining high mechanical performance silk fibers by feeding purified carbon nanotube/lignosulfonate composite to silkworms](#)

[Hao Xu](#), Wenhui Yi\*, Dongfan Li, Ping Zhang, Sweejiang Yoo, Lei Bai, Jin Hou, and Xun Hou  
(\* corresponding author)  
2019, **RSC Advances**, 9(7), 3558-3569.

[Supramolecular interactions of poly \[\(9, 9-dioctylfluorenyl-2, 7-diyl\)-co-thiophene\] with single-walled carbon nanotubes](#)

Ping Zhang, Wenhui Yi\*, [Hao Xu](#), Chao Gao, Jin Hou, Weiqiu Jin, Yue Lei, and Xun Hou  
2018, **Nanotechnology Reviews**, 7(6), 487-495.

[Recent advances in siRNA delivery for cancer therapy using smart nanocarriers](#)

Penghui Zhang, Keli An, Xiaoman Duan, [Hao Xu](#), Fei Li, Feng Xu\*  
2018, **Drug Discovery Today**, 23 (4), 900-911.

Other contribution proceedings:

[Scanning microwave microscopy for investigations of mechanical vibrations and mode coupling](#)

Xin Zhou\*, [Hao Xu](#), Didier Theron, Sophie Eliet, Christophe Boyaval, Pascal Tilmant, Francois Vaurette, Dmitri Yarekha, Jeanmarcel Houpin  
2023, *Frontiers of Nanomechanical Systems (FNS) workshop*, Delft, The Netherlands

[Scanning microwave microscopy for detecting mechanical vibrations of silicon nitride membranes](#)

[Hao Xu](#), Didier Theron, Sophie Eliet, Xin Zhou\*  
2022, *GDR seminar on Optomechanics and Nanomechanics*, Bordeaux, France

## TECHNICAL SKILLS

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Developer Languages: Python (Matplotlib, Numpy, Pandas, Scipy)

Developer Tools: Mathematica (Maple), COMSOL,  $\text{\LaTeX}$  (Overleaf), Origin, Github etc

## OUTSIDE OF RESEARCH

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Triathlon (ironman 70.3 race), Swimming, Marathon, Cooking